Missing Data

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach validation as a declared decision with inspectable evidence.

Supply complete scripts and row populations, distinguish policy from measurement and reserve the new count filter for assessment.

Keep one habit: Ask for raw value, exact type, bound, outcome, reason and coverage before accepting a report.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Detect absent and unusable values

Teach Week 14 as a two-session investigation using fictional data. A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60.

Keep rejected records visible for review instead of replacing their values with guesses. Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can validate **missing value**s without confusing zero with absent data.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Python 3 editor with a fresh local practice file
- Paper trace table and the student workbook
- Fictional **input** records from this lesson; no real student identifiers

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Noise and Outliers. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate mean and median

Reference example, not the only acceptable wording: Sum 30 divided by 5 gives 6.

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Look for: Explain why overall accuracy can hide failure on a less common class

Reference example, not the only acceptable wording: The **training**-majority rule always predicts clear. Matrix [[16,0],[4,0]] gives 16/20=80 percent, FP=0,FN=4.

It misses all four blocked cases, so the overall score hides complete failure on that evaluated class.

No. The first is text and the second is an integer.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Noise and Outliers

Choose Informative Questions

Accuracy as a Fraction

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Scores award 2 for preferred genre and 1 for short length. A book matches genre but page count is missing. Compare rejecting it, assigning zero length points and asking for pages.

Concept to explain

A system needs an explicit policy for unknown fields.

Term	Meaning
missing data	Information that was not recorded or is unavailable, which is different from an observed zero.
default	The value or behaviour used when no alternative is specified.
abstain	Decline to make a prediction when the available evidence does not meet the decision rule.
</details>

Term	Meaning
impute	Replace a missing value using a stated method; an imputed value is an estimate, not an observation.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the type-first condition and preserved review records

> Keep duration, stock and independent count contracts distinct.

State the duration schema and supply its complete script

A fictional activity log contains an absent duration, a recorded zero and an out-of-range value. Our schema accepts exact integer minutes from 0 through 60.

Keep rejected records visible for review instead of replacing their values with guesses.

```
rows = [{"minutes": 0}, {}, {"minutes": 12},
        {"minutes": -1}, {"minutes": None}]
```

```
accepted = []
rejected = 0
for row in rows:
    value = row.get("minutes")
    if type(value) is int and 0 <= value <= 60:
        accepted.append(value)
    else:
        rejected = rejected + 1
print(accepted)
print(rejected)
```

Explain get, append and exact type

dict.get returns None when this key is absent. An explicit None also represents unknown in this dataset. append adds an item to the end of a list. The script prints [0, 12] and then 3.

The exact type check prevents comparing None or text with the numeric range.

Collect the added 60/text 5 prediction

> **Add {"minutes": 60} and {"minutes": "5"}. Predict the accepted list and rejection count.**

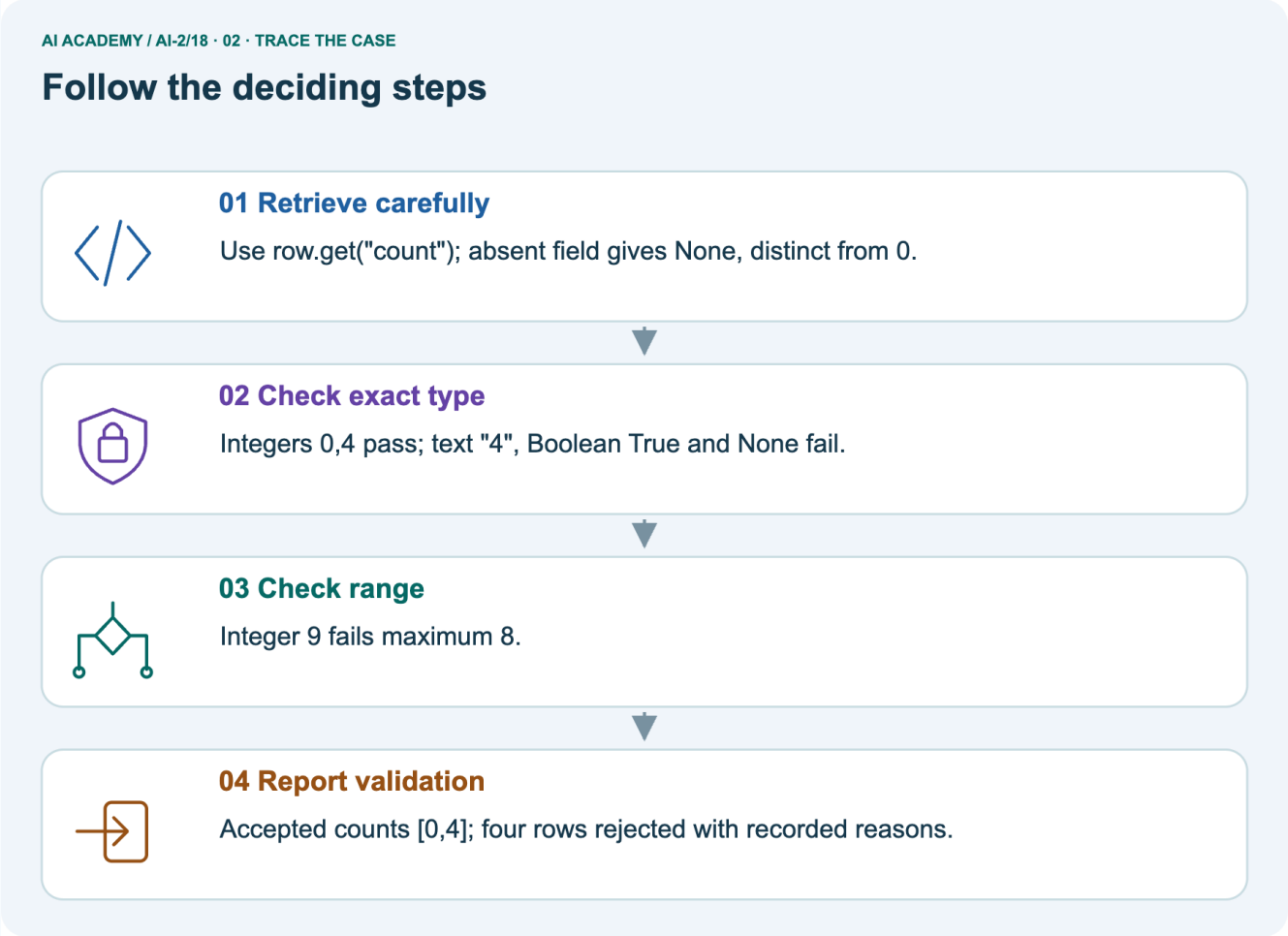
State the separate stock schema and all its rows

A fictional catalogue records counts of available copies under a strict **input** contract.

Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

Trace the Boolean and range decisions



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero.

Run the complete stock reason-recording script

```
rows = [{"count": 0}, {"count": 4}, {},
        {"count": "4"}, {"count": True}, {"count": 9}]
accepted = []
```

```
review = []
for number, row in enumerate(rows, 1):
    value = row.get("count")
    if value is None:
        reason = "unknown count"
    elif type(value) is not int:
        reason = "count is not an exact integer"
    elif not 0 <= value <= 8:
        reason = "count outside 0 through 8"
    else:
        accepted.append(value)
        continue
    review.append((number, row.copy(), reason))
print(accepted)
print(len(review))
```

Predict and check an authorised conversion

Change: Replace text "4" with integer 4 through an authorised conversion.

Prediction: Accepted values become [0,4,4]; three rows remain rejected.

Reason: The converted value now satisfies both type and range checks.

Limit the inference and elicit picture reasoning

Type and range checks cannot confirm whether the real shelf count was measured accurately.

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Reader's length preference is missing.

2. Ask: best when the user can answer.

3. **Abstain**: safest if the field is essential; **default**: convenient but may misrepresent.

4. Record which policy was used so the explanation stays truthful.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Genre is known, but page count is missing and length affects recommendation score.

Choose transparent handling.

Exact explanation to model

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Explain the added worked example

Key terms

Term	Meaning
Missing value	A value that was not recorded
Validation	Checking data against explicit rules
Rejection	Holding a record out because it fails a rule
Coverage	Which intended records or groups are represented

Model the reasoning aloud

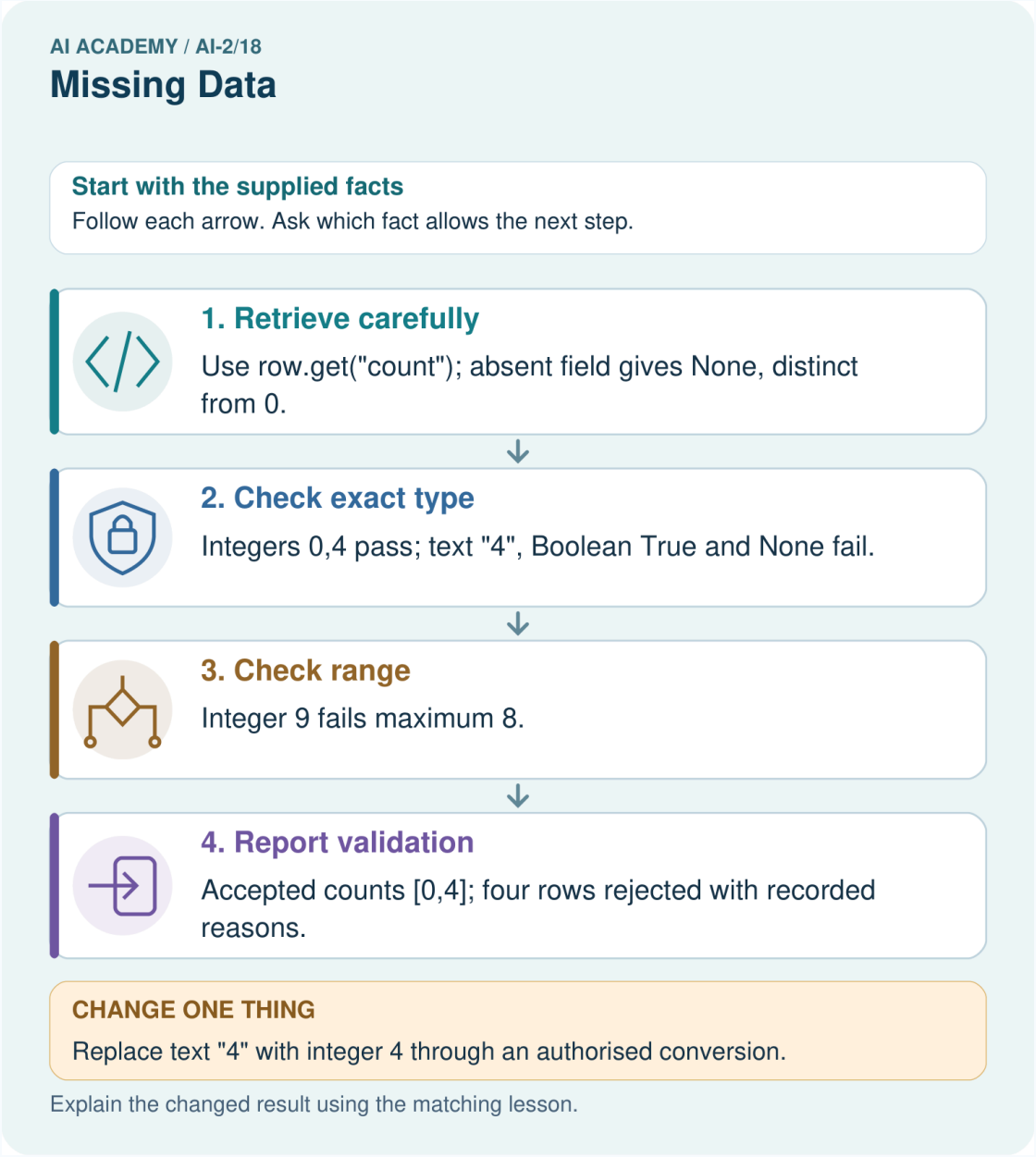
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Missing Data

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve carefully	Use row.get("count"); absent field gives None, distinct from 0.
2. Check exact type	Integers 0,4 pass; text "4", Boolean True and None fail.
3. Check range	Integer 9 fails maximum 8.
4. Report validation	Accepted counts [0,4]; four rows rejected with recorded reasons.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

Expected reasoning and fading help

Teacher check: Accepted [4, 0]; 2 rejected.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Missing value**s and **validation**. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Rejection** can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

The contract requires actual integer counts, not Boolean flags; type(True) is bool.

Changed-case reasoning: Accepted values become [0,4,4]; three rows remain rejected. The converted value now satisfies both type and range checks.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Genre is known, but page count is missing and length affects recommendation score.

Write the question to explain: Choose transparent handling.

Reveal after **prediction**s: Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Use the earlier changed question as optional later practice. Reserve Validate independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Accepted [0, 12, 60]; 4 rejected. Text "5" fails our rule. Exit: The summary excludes three sessions, which may differ from accepted ones. Report the exclusion and reasons rather than implying full **coverage**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Original duration script accepts[0,12],rejects3 of5. Adding60 and text"5" gives[0,12,60],rejects4 of7; addingTrue gives the same list,rejects5 of8. Extended review IDs are2unknown,4range,5unknown,7type,8type. Separate stock accepts[0,4],rejects4; authorised conversion of text4 adds another4 and leaves3 rejected. get alone does not distinguish absent versus explicit None, which are both unknown under this schema. Exact type rejects Booleans; neither validation nor a configured policy establishes truth of a measurement. Reserve the 0–10 independent task for W4.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the supplied extended eight-record duration audit for the main investigation. Retain accepted values and rejected row copies with reasons. The four explicitly supplied fictional recommender scenarios are an optional ask/**abstain**/fallback extension, not missing activity-log measurements.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A successful run proves the conclusion: **Rejection** can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: integer score 0 through 5 for 5, None, 0, 6 and True. Accept [5, 0]; reject 3. Exact type rejects Boolean True.

A convenient shortcut preserves meaning: Using if value would discard the legitimate zero and accept some invalid nonempty text. Test the written type and range rules explicitly.

Filling an absent duration with 0 falsely claims an observed zero.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Validate independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected independent count filter

> Keep this key until after the first individual attempt.

Selected W4 and checked key

Selected case: Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

Use get, exact int check and inclusive bounds. Accept [3, 0]; reject 3 records. Do not invent values for missing entries.

Judge implementation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension	Evidence required
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 zero/unknown key

Explain the difference between {"minutes": 0}, {} and {"minutes": None}.

0 is a recorded allowed duration. The other two are unknown under this schema.

W2 four-duration key

Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

Accepted [4, 0]; 2 rejected.

Process the four entries in order. The integers 4 and 0 have the required type and lie within 0 through 60, so keep [4, 0]. The integer 61 fails the upper boundary.

The text "4" fails the exact integer type, even though its characters look numeric. That makes two rejected entries.

W3 condition-repair key

A program uses if value to accept durations. Give a counterexample and the correct rule.

0 is wrongly rejected; nonempty text can be accepted. Require exact integer type and 0 <= value <= 60.

W5 coverage key

Only 2 of 5 supplied durations were accepted. Why should the report include this?

The summary excludes three sessions, which may differ from accepted ones. Report the exclusion and reasons rather than implying full **coverage**.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Retrieve or ask for the count, or use a documented partial-score policy with uncertainty. Do not silently insert zero.

Additional transfer practice question

A book's page count is blank. Is blank the same as zero? Give an honest way for a short-book recommender to respond.

Reasoned practice answer

Blank means unknown, not zero. Ask for the page count or mark length unknown and avoid claiming it meets the short-book rule.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: "**Missing data** means the numerical value is zero." Keep the first attempt and an uncoached retry separate.

Original transfer answer

Rejecting may lose a good candidate; zero points treats unknown as no match and must be disclosed; asking can resolve uncertainty. A partial score is 2 with length unknown, not a confirmed complete score of 2.

Choose a policy before evaluation.

Explanations for the additional workbook examples

W1 Distinguish zero and unknown

W2 Trace filtering

W3 Repair a condition

W4 Validate independently

Write a filter for count values that must be integers from 0 to 10. **Process** {count: 3}, {}, {count: 0}, {count: 11}, {count: "3"}, using quoted keys in Python. Report accepted values and rejected count.

Use get, exact int check and inclusive bounds. Accept [3, 0]; reject 3 records. Do not invent values for missing entries.

W5 Report the limitation

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Use ‘no preference’ only when the reader confirms it; otherwise ask or explain uncertainty.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use three piles: accepted zero, accepted other and review. Extension: retain each rejected row with a reason instead of just counting it.

Independent W4 **evidence**: Use get, exact int check and inclusive bounds. Accept [3, 0]; reject 3 records. Do not invent values for missing entries. Record whether the learner needed no help, a hint or explanation.

Fresh retry: integer score 0 through 5 for 5, None, 0, 6 and True. Accept [5, 0]; reject 3. Exact type rejects Boolean True.

Plan the delayed check

Suggested return points in this level: ai-2/19, ai-2/21, ai-2/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 19: Sampling and Representation. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 19: Sampling and Representation. Carry forward supplied, accepted and reviewed counts; filtering can change which cases remain represented even when every retained value passes its contract.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Detect absent and unusable values

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

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> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Predict accepted values and **rejection**s for minutes 4, 0, 61 and "4".

Reasoned practice answer: Accepted [4, 0]; 2 rejected.

Check every learner before discussion: Use the worked case for **Missing value**s and **validation**. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Rejection** can change which sessions remain represented. Report 2 accepted of 5 supplied here. A summary of accepted rows does not automatically describe the missing or rejected sessions.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace text "4" with integer 4 through an authorised conversion. Require a **prediction** and a reason before comparing. Expected result: Accepted values become [0,4,4]; three rows remain rejected.

Reason: The converted value now satisfies both type and range checks.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

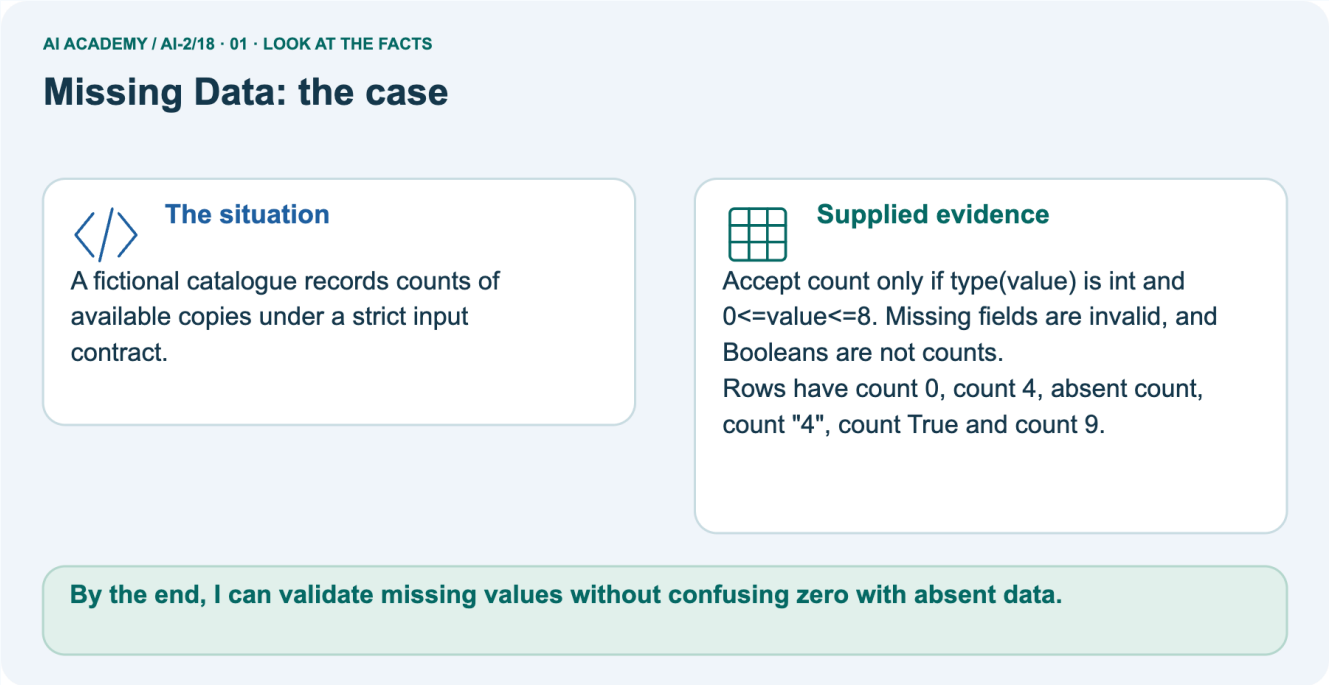
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Missing Data: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

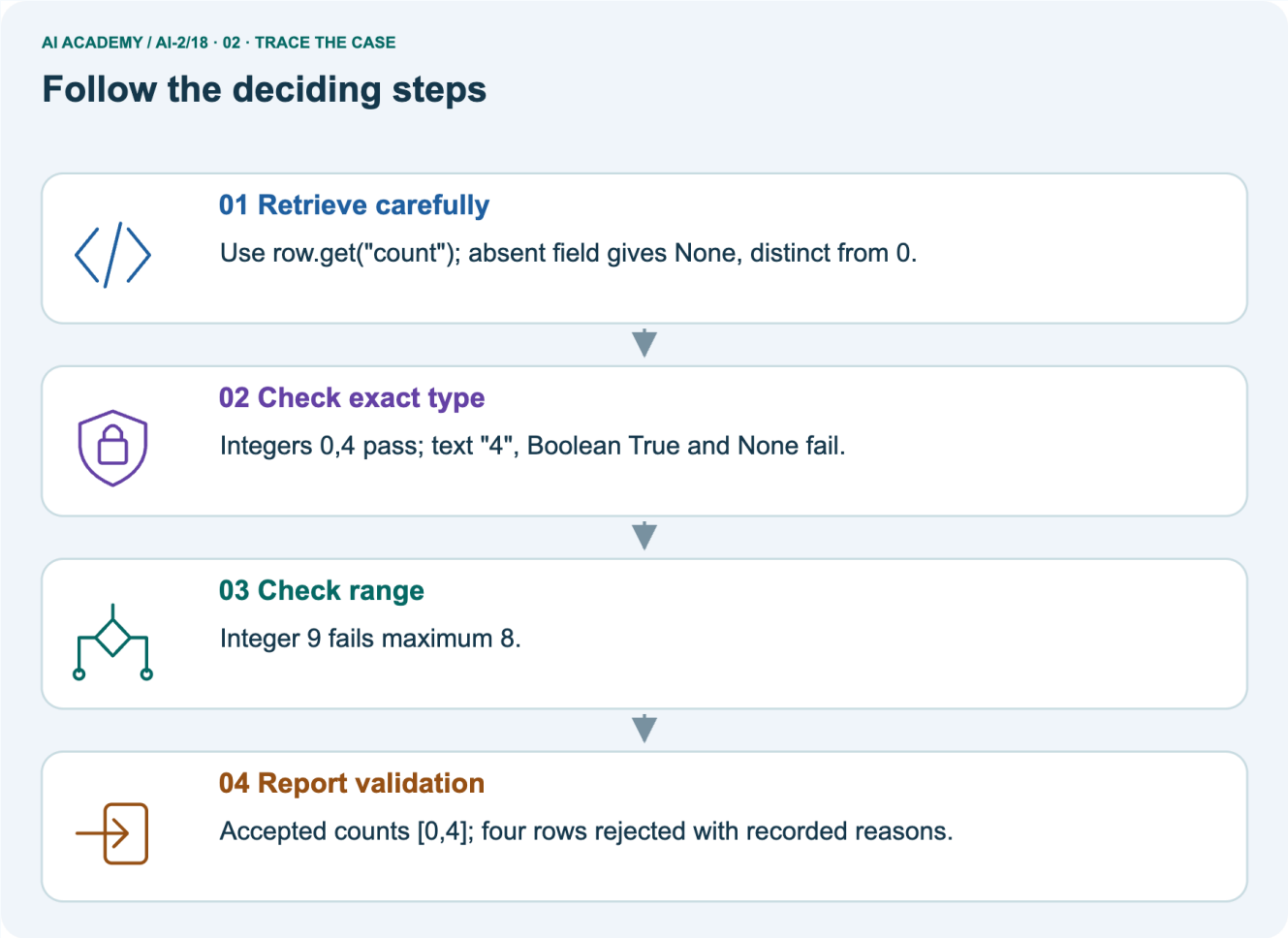
A fictional catalogue records counts of available copies under a strict **input** contract. Supplied facts: Accept count only if type(value) is int and 0<=value<=8. Missing fields are invalid, and Booleans are not counts.

Rows have count 0, count 4, absent count, count "4", count True and count 9.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



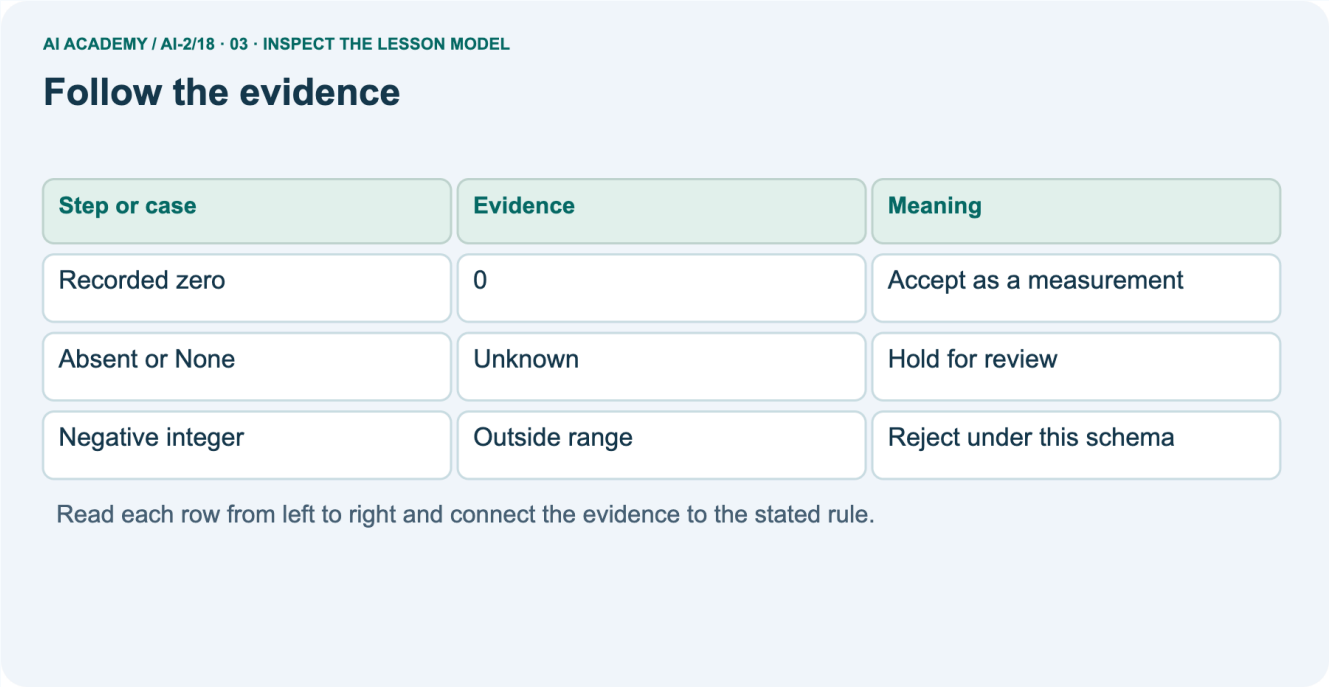
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve carefully: Use row.get("count"); absent field gives None, distinct from 0. → Check exact type: Integers 0,4 pass; text "4", Boolean True and None fail. → Check range: Integer 9 fails maximum 8. → Report **validation**: Accepted counts [0,4]; four rows rejected with recorded reasons.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



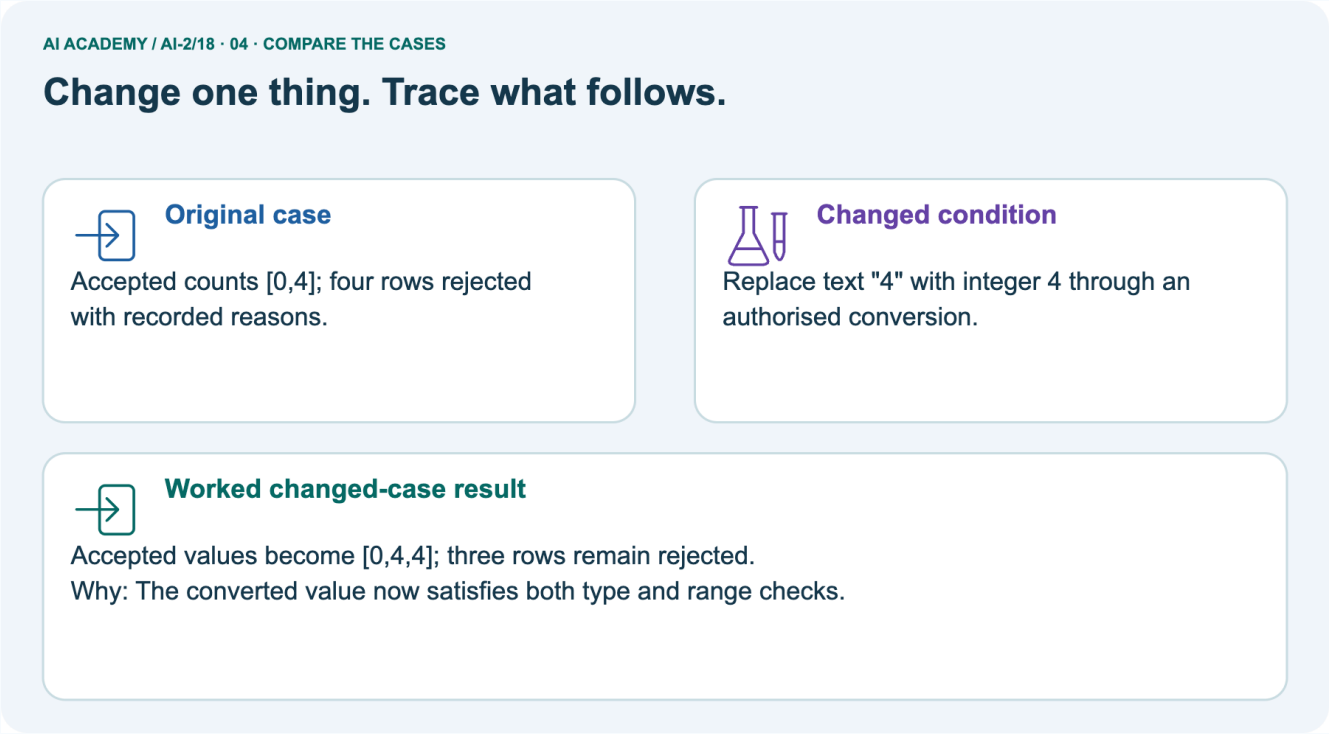
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Recorded zero | 0 | Accept as a measurement. Absent or None | Unknown | Hold for review. Negative integer | Outside range | Reject under this schema

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accepted counts [0,4]; four rows rejected with recorded reasons. Changed condition: Replace text "4" with integer 4 through an authorised conversion.

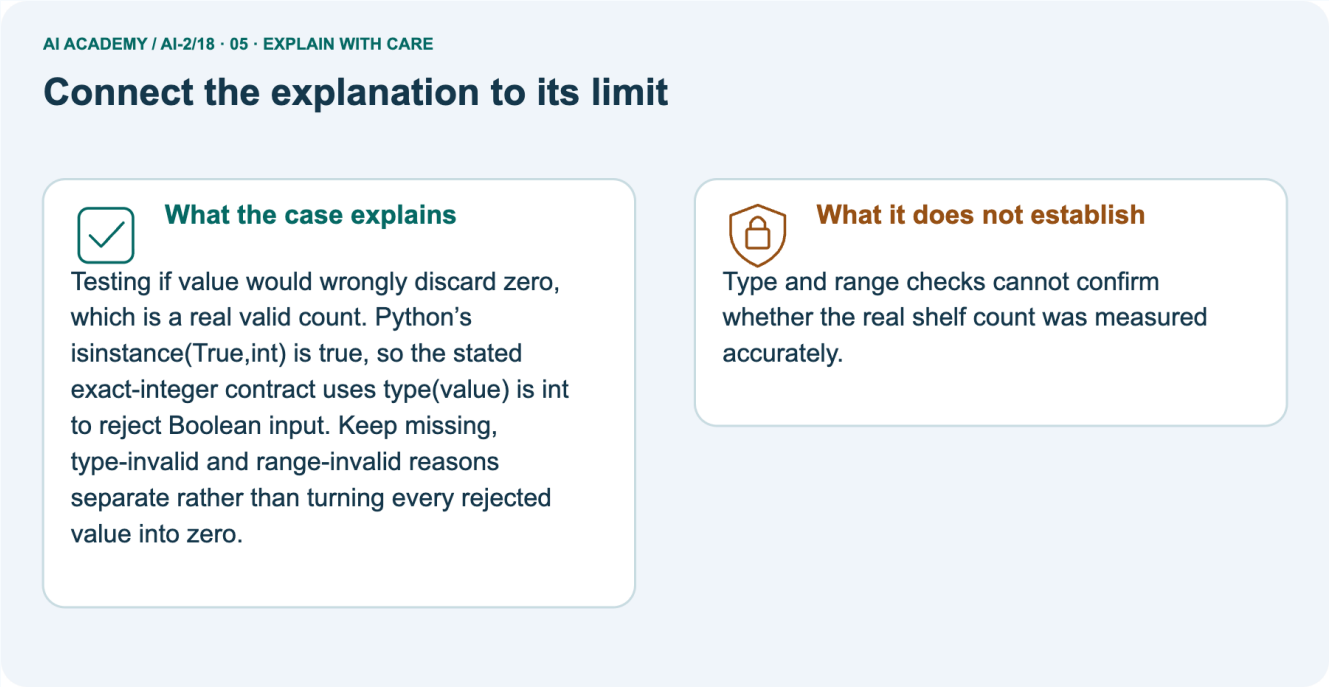
Expected result: Accepted values become [0,4,4]; three rows remain rejected. Reason: The converted value now satisfies both type and range checks.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The converted value now satisfies both type and range checks.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Testing if value would wrongly discard zero, which is a real valid count. Python's isinstance(True,int) is true, so the stated exact-integer contract uses type(value) is int to reject Boolean **input**.

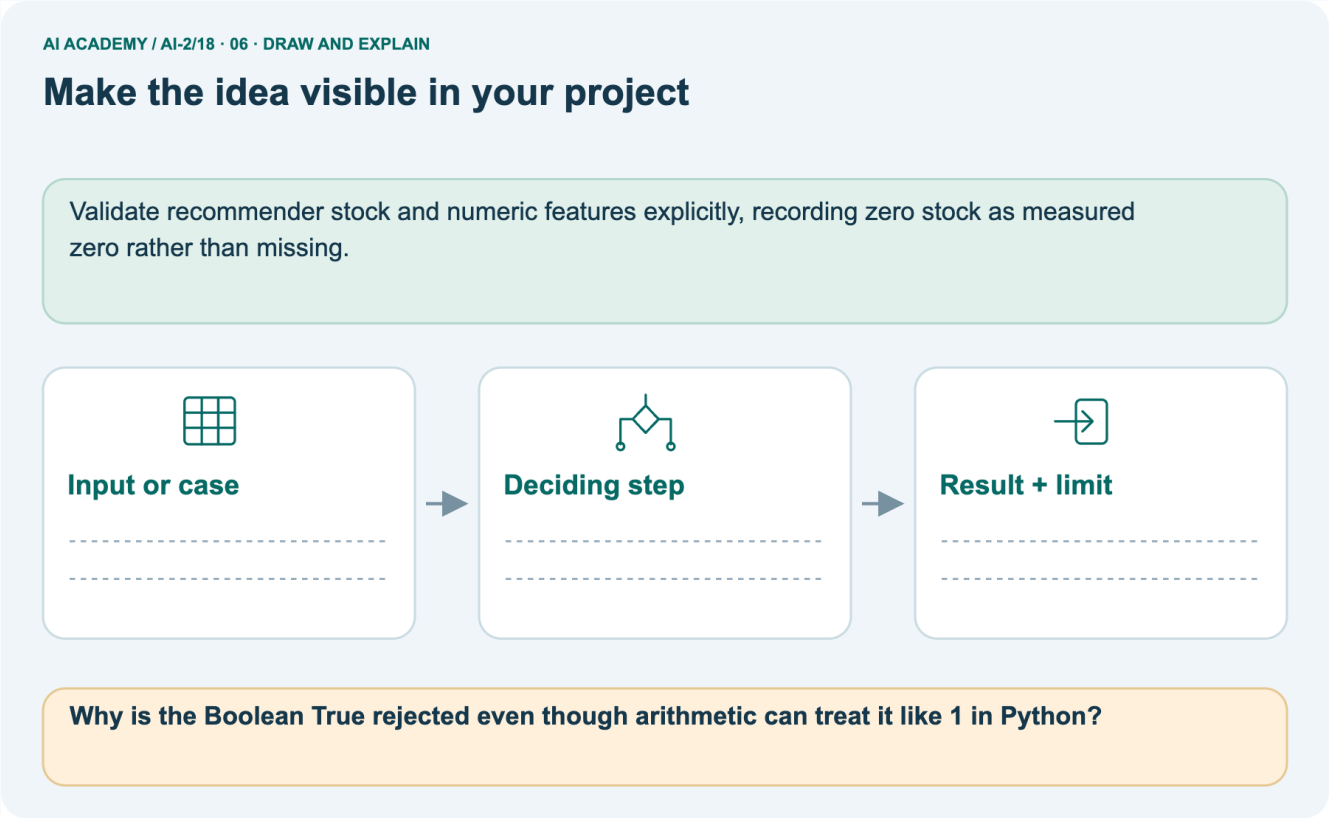
Keep missing, type-invalid and range-invalid reasons separate rather than turning every rejected value into zero. Limit: Type and range checks cannot confirm whether the real shelf count was measured accurately.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Type and range checks cannot confirm whether the real shelf count was measured accurately.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender stock and numeric features explicitly, recording zero stock as measured zero rather than missing. Draw your **input**, deciding step, result and limit.

Supported question: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The contract requires actual integer counts, not Boolean flags; type(True) is bool.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Missing Data: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.